

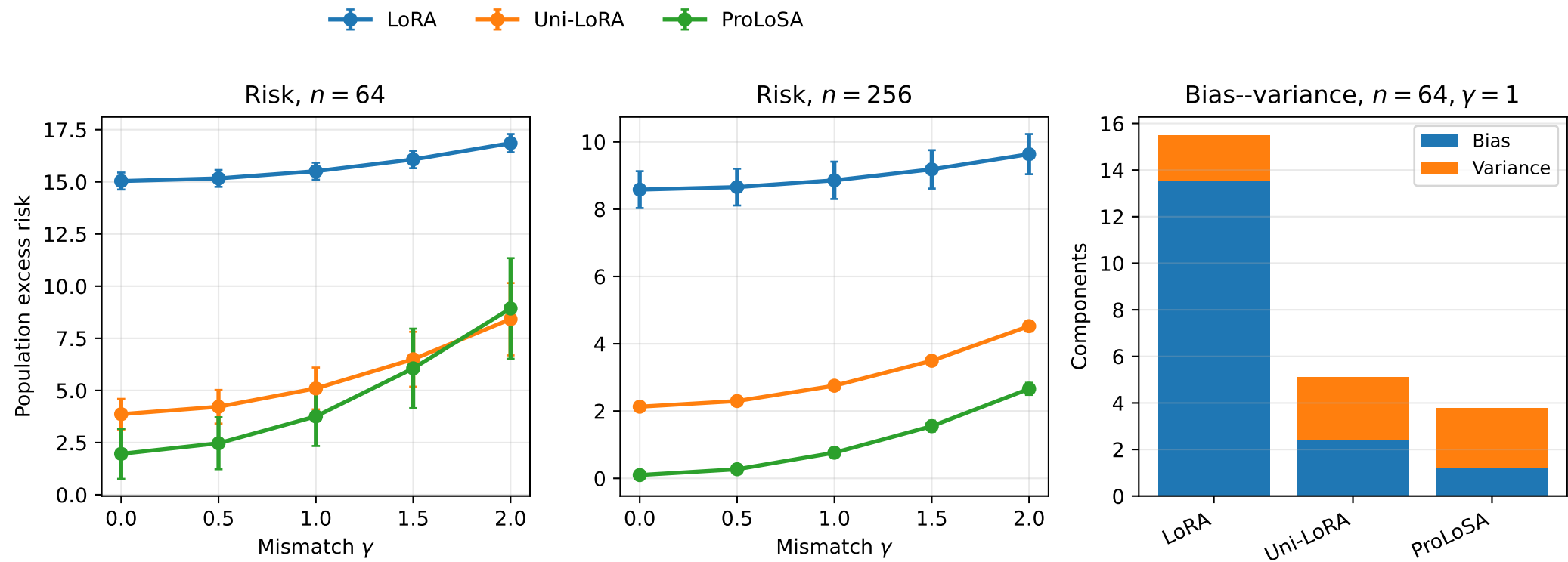


Figure S1: Known-projection synthetic validation of the bias-variance theory. Left and middle: population excess risk as subspace mismatch γ increases under different sample sizes. Right: empirical bias-variance decomposition at $n=64$ and $\gamma=1$. ProLoSA uses warmup SNIP scores to select sparse residual coordinates.